



VITAMINS & MINERALS

complete guide

Disclaimer

The information contained in this ebook is for educational and informational purposes only and is not intended as medical advice. It is not intended to diagnose, treat, cure, or prevent any disease, nor should it be used as a substitute for professional medical advice, diagnosis, or treatment. Always seek the advice of your physician or another qualified healthcare provider with any questions you may have regarding a medical condition.

The nutritional information provided in this ebook is based on current scientific understanding and is subject to change as new research becomes available. The authors and publisher make no representation or warranties concerning the accuracy or completeness of the ebook's contents and specifically disclaim any implied warranties of merchantability or fitness for any particular purpose.

The authors and publisher shall have no liability or responsibility to any person or entity with respect to any loss or damage caused or alleged to be caused directly or indirectly by the information contained in this ebook. Any reliance you place on such information is therefore strictly at your own risk.

The opinions expressed in this ebook are those of the authors and do not necessarily reflect the views of the publisher. The publisher does not endorse any specific product or service mentioned in this ebook.

All trademarks, service marks, trade names, logos, and graphics used in this ebook are the property of their respective owners.



Welcome!

Welcome to Enchanted's Weight Loss and Wellness Program!
We hope you find the information in this guide helpful during your journey to a healthier lifestyle.

DR. BRIDGET MARTIN

DNP, APRN, NP-C

ENCHANTED MEDICAL AESTHETICS



TABLE OF CONTENTS

Vitamin A	6
Vitamin C	9
Vitamin D	12
Vitamin E	16
Vitamin K	19
B vitamins	22
Calcium	24
Iron	27
Manesium	30
Zinc	33
Sources & Further Reading	36
Common Myths and Misconceptions	38

VITAMIN A

Vitamin A is a fat-soluble nutrient that is essential for maintaining healthy vision, skin, and immune function. It is a group of compounds that includes retinol, retinal, and retinoic acid.

One of the most well-known functions of vitamin A is its role in promoting good vision. It is a critical component of the pigment rhodopsin, which is found in the cells of the retina and is necessary for vision in low-light conditions. Vitamin A also plays a role in the growth and development of many different types of cells in the body, including skin cells and immune cells.

In addition to its importance for vision and immune function, vitamin A also helps to maintain healthy skin, as it is involved in the production and turnover of skin cells. It is often found in skincare products due to its ability to improve skin texture and reduce the appearance of fine lines and wrinkles.

Some studies suggest that vitamin A may help prevent certain types of cancer, particularly lung cancer in smokers. However, high doses of vitamin A can be toxic and may increase the risk of cancer.

Vitamin A is important for fetal development, particularly for the development of the eyes, and a deficiency in vitamin A during pregnancy can lead to birth defects. However, excessive intake of vitamin A during pregnancy can also be harmful to the fetus.

Overall, vitamin A is an important nutrient that plays a variety of roles in the body. While a deficiency in vitamin A can be harmful, excessive intake of vitamin A can also have negative health effects, so it is important to obtain an appropriate amount through a balanced diet.



The 5 best foods for vitamin A, that should be on your shopping list:

1) Beef liver 	Animal livers are among the richest sources of vitamin A. This is because, like humans, animals store vitamin A in the liver. A 3-ounce (oz) serving of pan-fried beef liver contains 6,582 micrograms (mcg) of vitamin A, which equates to 731% of the daily value (DV). The DV allows people to compare the nutrient contents of different foods easily. It is a percentage based on the recommended daily intake of key nutrients from the Food and Drug Administration (FDA).
2) Fish liver 	Fish livers are also excellent sources of preformed vitamin A, with 1 tablespoon of cod liver oil providing 4,080 mcg. This and other fish oils are among the richest sources of omega-3 fatty acids, which help fight inflammation and protect the heart. Research also suggests that they may treat or prevent depression.
3) Sweet potato 	One whole sweet potato, baked in its skin, provides 1,403 mcg of vitamin A, which is 156% of the DV. The vitamin A present in this root vegetable is in the form of beta-carotene. Research suggests that this compound may help protect against age-related macular degeneration (AMD).
4) Carrots 	Carrots are rich in beta-carotene. Half a cup of raw carrots contains 459 mcg of vitamin A and 51% of the DV. A large carrot contains around 29 calories. This makes for a light and healthful snack, especially when eaten with hummus or guacamole.
5) Black-eyed peas 	Beans are an excellent source of plant-based protein, and they are also rich in fiber. Each cup of boiled black-eyed peas contains 66 mcg of vitamin A and 7% of the DV. Studies support the role of various types of beans in promoting heart health. For instance, research has linked eating beans with a lower risk of heart disease and high blood pressure.

DV = Daily Value
IU = International Unit

How much vitamin A do we need?

The United States Department of Agriculture (USDA) provides dietary guidelines for Americans, which include recommendations for vitamin A intake. The current guidelines for vitamin A and other related specifications are shown in the table.

The recommended daily intake of vitamin A for adults is 700-900 micrograms of Retinol Activity Equivalents (RAE). Pregnant and breastfeeding women have higher requirements, with a recommended daily intake of 770-1300 micrograms RAE.

RAE stands for Retinol Activity Equivalents. It is a measure of the amount of vitamin A in a food or supplement that takes into account the different forms of vitamin A and their varying biological activities. Retinol is the most active form of vitamin A, while other forms such as beta-carotene have less activity. RAE is used to standardize and compare the vitamin A content of different foods and supplements based on their retinol activity. One microgram of retinol is equivalent to 1 RAE, while one microgram of beta-carotene from food is equivalent to 0.3 RAE, and one microgram of beta-carotene from supplements is equivalent to 0.15 RAE.

Life stage	RDA
Birth to 6 months	400 mcg
Infants 7–12 months	500 mcg
Children 1–3 years	300 mcg
Children 4–8 years	400 mcg
Children 9–13 years	600 mcg
Teens 14–18 years (boys)	900 mcg
Teens 14–18 years (girls)	700 mcg
Adults (men)	900 mcg
Adults (women)	700 mcg
Pregnant women	770-1300mcg
Breastfeeding women	1300-1500 mcg



VITAMIN C

Vitamin C, also known as ascorbic acid, is a water-soluble vitamin that is essential for maintaining good health. It is not produced by the human body, so it must be obtained through diet or supplements.

Vitamin C has several important functions in the body. It acts as an antioxidant, protecting cells from damage caused by free radicals. It also helps to boost the immune system, supporting the body's ability to fight off infections and diseases. Vitamin C is involved in the production of collagen, a protein that is essential for the growth and repair of tissues in the body. It also helps the body to absorb iron from plant-based foods, which is important for the production of red blood cells.

A deficiency of vitamin C can lead to a condition called scurvy, which is characterized by fatigue, muscle weakness, joint and muscle aches, and skin rashes. However, vitamin C deficiency is rare in developed countries, as it is found in many foods and added to some fortified foods.



The 5 best foods for vitamin C, that should be on your shopping list:

1) Citrus fruits 	Citrus fruits like oranges, grapefruit, lemons, and limes are well-known for their high vitamin C content. For example, one medium-sized orange contains about 70 milligrams of vitamin C, which is more than the daily recommended intake for adults.
2) Bell pepper 	Red, yellow, and green bell peppers are all good sources of vitamin C. For example, one medium-sized red bell pepper contains about 150 milligrams of vitamin C, which is more than twice the daily recommended intake for adults.
3) Broccoli 	Broccoli is a cruciferous vegetable that is high in many vitamins and minerals, including vitamin C. One cup of cooked broccoli contains about 80 milligrams of vitamin C.
4) Strawberries 	Strawberries are a delicious and nutritious fruit that are also high in vitamin C. One cup of sliced strawberries contains about 85 milligrams of vitamin C, along with other beneficial nutrients like fiber and antioxidants.
5) Guava 	Guava is a very good source of vitamin C. In fact, guava is one of the richest sources of vitamin C among all fruits. One cup of fresh guava contains about 377 milligrams of vitamin C, which is more than four times the daily recommended intake for adults. Guava also contains other beneficial nutrients like fiber, potassium, and antioxidants.

DV = Daily Value
IU = International Unit

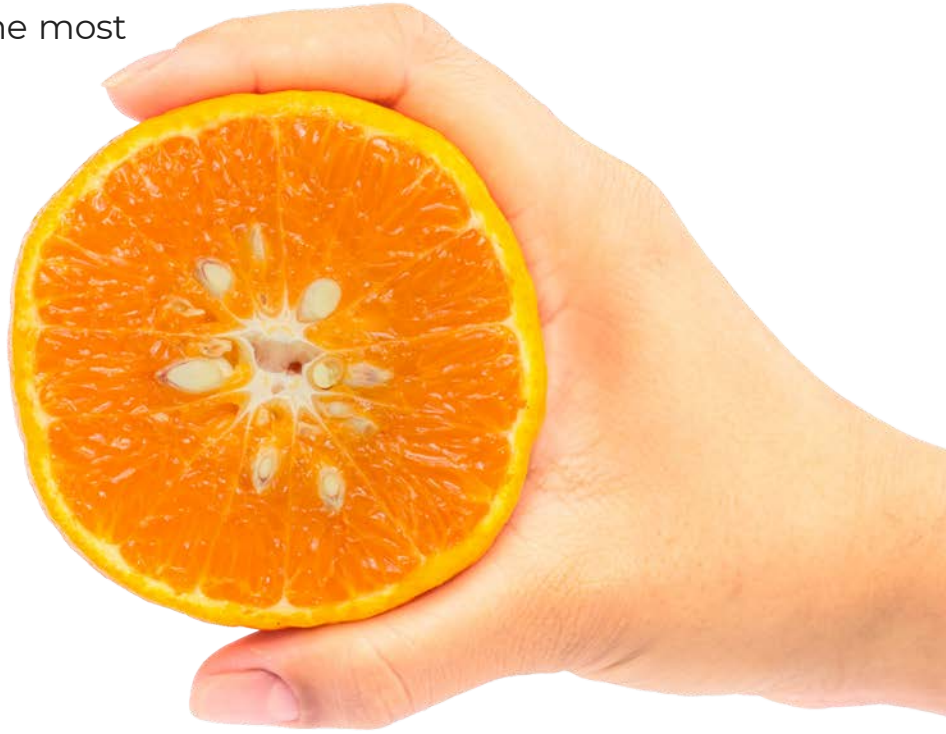
How much vitamin C do we need?

The Recommended Dietary Allowance (RDA) for Vitamin C varies by age, gender, and physiological condition, such as pregnancy and breastfeeding. The current guidelines for vitamin C and other related specifications are shown in the table.

It's worth noting that smokers and those exposed to secondhand smoke require higher amounts of Vitamin C due to the oxidative stress caused by smoking. They should consume an additional 35 mg per day compared to the above RDA values.

Cooking may reduce the amount of vitamin C in fruits and vegetables. The National Institutes of Health Office of Dietary Supplements (ODS) notes that steaming or microwaving these foods may retain most of the vitamins. People should eat various raw fruits and vegetables every day to obtain the most vitamin C.

Life stage	RDA
Birth to 6 months	40 mcg
Infants 7–12 months	50 mcg
Children 1–3 years	15 mcg
Children 4–8 years	25 mcg
Children 9–13 years	45 mcg
Teens 14–18 years (boys)	75 mcg
Teens 14–18 years (girls)	65 mcg
Adults (men)	90 mcg
Adults (women)	75 mcg
Pregnant women	85 mcg
Breastfeeding women	120 mcg



VITAMIN D

Vitamin D is a micronutrient that is needed for optimal health throughout your whole life.

It is a fat-soluble vitamin, meaning that it dissolves in fat/oil and can be stored in the body for long periods of time. There are two main forms, D2 and D3; D3 being more effective.

Vitamin D is also unique, because it is both a nutrient we eat, and a hormone that our body makes from exposure to sunlight, which is why it is often named the 'sunshine' vitamin.

While vitamin D affects various cells related to bone health –like telling the cells in the gut to absorb calcium and phosphorus –but scientists have now generated a strong body of evidence supporting vitamin D as an indispensable hormone, required for regulation of many physiologic functions.

Research shows that vitamin D receptors are present in nearly every tissue and cell in the body, and adequate vitamin D is essential for optimal function of these tissues and cells to carry out important roles, including: reduction of inflammation, modulation of cell growth, immune function and glucose metabolism.

Simply put: Sufficient levels of vitamin D are essential for disease prevention, longevity, and optimizing human health.



The 5 best foods for vitamin D, that should be on your shopping list:

1) Salmon 	According to the United States Department of Agriculture (USDA) Food Composition Database, one 3.5-ounce (100-gram) serving of farmed Atlantic salmon contains 526 IU of vitamin D, or 66% of the DV. However, wild-caught salmon can contain 988 IU of vitamin D per 3.5-ounce (100-gram) serving, or 124% of the DV. Ideally aim for wild-caught, but either way, salmon is an easy fatty fish to get your source of vitamin D.
2) Herring 	Herring feed on small fish such as anchovies and sardines, as well as small weed-dwelling crustaceans and insects. Fresh Atlantic herring provides 216 IU per 3.5-ounce (100-gram) serving, which is 27% of the DV.
3) Sardines 	Wild caught ocean-raised fish, such as sardines, feed on vitamin D–rich plankton, so they contain much higher levels of vitamin D than farm-raised varieties of fish. One can of sardines (3.8 ounces) contains on average 177 IU, or 22% of the DV.
4) Eggs (yolks) 	Most of the protein in an egg is found in the white. However, fat, vitamins, and minerals are found mostly in the yolk. One typical egg yolk contains 37 IU of vitamin D or 5% of the DV. Choose carefully because vitamin D levels in chicken egg yolk depend largely on sun exposure. Pasture-raised chickens that roam outside in the sunlight can produce eggs with higher levels of vitamin D.
5) Mushrooms 	Mushrooms are the only non-animal, unfortified food source of vitamin D that can provide a substantial amount of D2 in a single serve. Some brands grow mushrooms in the dark, which drastically reduces vitamin D2. While some brands use ultraviolet light, which can provide 130–450 IU of vitamin D2 per 3.5 ounces (100 grams). Choose only wild or sun-dried mushrooms or mushrooms treated with UV light for a good source of vitamin D.

DV = Daily Value
IU = International Unit

Fortified Foods

Natural sources of vitamin D are limited, especially if you're vegetarian or plant-based. Fortunately, over the past few decades, many foods have been fortified with vitamin D.

These can include:

	Tofu		Soy milk
	Cereal		Orange juice
	Cheese		Oatmeal
	Cow’s milk		Yogurt

How much vitamin D do we need?

According to the USDA, the recommended daily allowances (RDA) of vitamin D for different age groups and genders are shown in the table.

It's worth noting that some health organizations recommend higher daily intakes of vitamin D for specific populations, such as those with limited sun exposure, dark skin, or certain medical conditions. It's always a good idea to speak with your healthcare provider to determine the appropriate daily intake for your needs.

**IU stands for "International Unit." It is a unit of measurement used to quantify the biological activity of vitamins, hormones, and other substances. For example, when measuring vitamin D, IU is used to express the potency of the vitamin. The amount of vitamin D needed to produce a certain biological effect may vary depending on the form of vitamin D, so IU allows for standardization across different forms.*

Life stage	RDA
Birth to 6 months	400 IU
Infants 7–12 months	400-600 IU
Children 1–3 years	600 IU
Children 4–8 years	600 IU
Children 9–13 years	600 IU
Teens 14–18 years (boys)	600 IU
Teens 14–18 years (girls)	600 IU
Adults (men)	600-800 IU
Adults (women)	600-800 IU
Pregnant women	600-800 IU
Breastfeeding women	600-800 IU



VITAMIN E

Vitamin E is a fat-soluble vitamin that functions as an antioxidant in the body. It occurs naturally in eight different forms, including alpha-tocopherol, which is the most biologically active form of the vitamin.

Vitamin E helps to protect the body's cells from damage caused by free radicals, which are unstable molecules that can damage cell membranes, DNA, and other important cellular structures. In addition to its antioxidant properties, vitamin E is involved in immune function, DNA repair, and other physiological processes.

There are eight distinct forms of vitamin E, but researchers believe that only one type, alpha-tocopherol, helps meet human nutritional needs.
Plenty of foods contain vitamin E, which means many people get enough of the vitamin naturally through their diet.

According to research from 2015, vitamin E can also improve skin health by reducing collagen breakdown and free radical damage in the skin.

Aside from its role as an antioxidant, vitamin E also helps support the immune system.



The 5 best foods for vitamin E, that should be on your shopping list:

1) Sunflower seeds 	Sunflower seeds make an excellent snack. People can also sprinkle them on yogurt, oatmeal, or salad. A 100-gram (g) serving of sunflower seeds contains 35.17 milligrams (mg) of vitamin E. Sunflower seeds are packed with a variety of nutrients and can help a person get enough fiber to keep their digestive system healthy.
2) Almonds 	For every 100 g serving of almonds, there is 25.63 mg of vitamin E. People can snack on roasted almonds, add them to cereal and baked goods, or drink almond milk. Almonds also contain 21.15 g protein, 12.5 g fiber, 733 mg potassium, 270 mg magnesium.
3) Avocado 	Avocados are a versatile fruit that contain very little sugar and plenty of nutrients. In 100 g of avocado, there is 2.07 mg of vitamin E. The same size serving also contains 10 mg of vitamin C, making it a healthful addition to many meals and snacks. Avocado also contains more potassium than bananas.
4) Spinach 	Spinach is a good source of vitamin E, but it is not considered a particularly rich source. One cup of cooked spinach contains approximately 0.6 mg of vitamin E, which is about 4% of the daily recommended intake for adults.
5) Butternut squash 	Butternut squash is a tasty vegetable common in many fall and winter dishes. There is 1.29 mg of vitamin E in 100 g of baked butternut squash.

How much vitamin E do we need?

The Recommended Dietary Allowance (RDA) for Vitamin E varies by age, gender, and physiological condition such as pregnancy and breastfeeding.

It's important to note that taking high doses of Vitamin E supplements can increase the risk of bleeding, so it is generally recommended to meet the daily requirements through a balanced diet.

**IU stands for "International Unit." It is a unit of measurement used to quantify the biological activity of vitamins, hormones, and other substances. For example, when measuring vitamin D, IU is used to express the potency of the vitamin. The amount of vitamin D needed to produce a certain biological effect may vary depending on the form of vitamin D, so IU allows for standardization across different forms.*

Life stage	RDA
Birth to 6 months	400 IU
Infants 7–12 months	400-600 IU
Children 1–3 years	600 IU
Children 4–8 years	600 IU
Children 9–13 years	600 IU
Teens 14–18 years (boys)	600 IU
Teens 14–18 years (girls)	600 IU
Adults (men)	600-800 IU
Adults (women)	600-800 IU
Pregnant women	600-800 IU
Breastfeeding women	600-800 IU



VITAMIN K

Vitamin K is a fat-soluble vitamin that is essential for blood clotting and bone health. It is also involved in other important biological processes, such as regulating calcium in the body and promoting proper growth and development.

There are two main forms of vitamin K: K1 (phylloquinone), which is found in leafy green vegetables and other plant sources, and K2 (menaquinone), which is produced by bacteria in the gut and found in animal products and fermented foods.

In the body, vitamin K is necessary for the production of certain proteins that help with blood clotting. It also plays a role in bone metabolism, helping to maintain bone density and prevent osteoporosis. Additionally, research has suggested that vitamin K may have other health benefits, such as reducing the risk of certain cancers and heart disease.

Deficiency in vitamin K can lead to an increased risk of bleeding and bruising, as well as weakened bones. However, deficiency is relatively rare, and most people can get enough vitamin K through a balanced diet that includes green leafy vegetables, such as kale, spinach, and broccoli, as well as other sources like dairy products, meats, and eggs.



The 5 best foods for vitamin K, that should be on your shopping list:

1) Kale 	One cup of raw kale provides over 600% of the daily recommended intake of vitamin K. You can add it to smoothies, and salads, or make kale chips as a healthy snack. 1/2 cup: 531 mcg (443% of the DV) 100 grams: 817 mcg (681% of the DV)
2) Spinach 	One cup of cooked spinach provides over 100% of the daily recommended intake of vitamin K. It's a versatile leafy green that can be used in salads, omelets, or soups. 1 cup: 145 mcg (121% of the DV) 100 grams: 483 mcg (402% of the DV)
3) Broccoli 	One cup of cooked broccoli provides about 220% of the daily recommended intake of vitamin K. It can be roasted, steamed, or added to stir-fries. 1/2 cup: 110 mcg (92% of the DV) 100 grams: 141 mcg (118% of the DV)
4) Brussels sprouts 	One cup of cooked Brussels sprouts provides about 170% of the daily recommended intake of vitamin K. They can be roasted, sautéed, or shaved raw into salads. 1/2 cup: 109 mcg (91% of the DV) 100 grams: 140 mcg (117% of the DV)
5) Collard greens 	One cup of cooked collard greens provides over 100% of the daily recommended intake of vitamin K. They can be used in soups, stews, or sautéed as a side dish. 1/2 cup: 386 mcg (322% of the DV) 100 grams: 407 mcg (339% of the DV)

How much vitamin K do we need?

Intake recommendations for vitamin K and other nutrients are provided in the Dietary Reference Intakes (DRIs) developed by the Food and Nutrition Board (FNB) at the Institute of Medicine of the National Academies. DRI is the general term for a set of reference values used for planning and assessing nutrient intakes of healthy people.

Insufficient data were available to establish an EAR for vitamin K, so the FNB established AIs for all ages that are based on vitamin K intakes in healthy population groups. The table lists the current AIs for vitamin K in micrograms (mcg). The AIs for infants are based on the calculated mean vitamin K intake of healthy breastfed infants and the assumption that infants receive prophylactic vitamin K at birth as recommended by American and Canadian pediatric societies

**Adequate intake not established, but breast milk or formula should provide enough vitamin K.*

Life stage	RDA
Birth to 6 months	*
Infants 7–12 months	2.5 mcg
Children 1–3 years	30 mcg
Children 4–8 years	55 mcg
Children 9–13 years	60 mcg
Teens 14–18 years (boys)	75 mcg
Teens 14–18 years (girls)	60 mcg
Adults (men)	120 mcg
Adults (women)	90 mcg
Pregnant women	90 mcg
Breastfeeding women	90 mcg



B VITAMINS

B vitamins are essential for maintaining good health and well-being. They are water-soluble vitamins, which means that they dissolve in water and are not stored in the body to a great extent. As a result, it is important to consume B vitamins on a regular basis to maintain optimal health.

- There are eight B vitamins, each with a unique function in the body. Thiamin (B1) plays a key role in energy metabolism and the functioning of the nervous system.
- Riboflavin (B2) is important for maintaining healthy skin, eyes, and mucous membranes.
- Niacin (B3) helps the body convert food into energy and supports the digestive and nervous systems.
- Pantothenic acid (B5) is involved in the production of hormones and cholesterol and is also important for healthy skin and hair.
- Pyridoxine (B6) is involved in over 100 enzyme reactions in the body, including the metabolism of protein, carbohydrates, and fats.
- Biotin (B7) helps the body break down and use protein, fats, and carbohydrates and is also important for healthy hair, skin, and nails.
- Folate (B9) is important for proper fetal development during pregnancy and also plays a role in the production of red blood cells.
- Finally, cobalamin (B12) is important for the production of red blood cells and also supports the nervous system.

Deficiencies in B vitamins can lead to a variety of health problems, including anemia, nerve damage, and birth defects. However, consuming too much of certain B vitamins, such as niacin and B6, can also have adverse effects on health. Therefore, it is important to consult with a healthcare provider before taking B vitamin supplements, especially at high doses.



The 5 best foods for B vitamins, that should be on your shopping list:

1) Salmon



This all-around nutritious fish is high in several B vitamins. In fact, a 3.5-ounce (oz), or 100-gram (g), cooked serving of salmon contains:

- Thiamine (B1): 23% of the DV
- Riboflavin (B2): 37% of the DV
- Niacin (B3): 63% of the DV
- Pantothenic acid (B5): 38% of the DV
- Pyridoxine (B6): 56% of the DV
- Cobalamin (B12): 127% of the DV

2) Leafy greens



Several leafy greens stand out for their folate (B9) content. These are among the highest vegetable sources of folate):

- Spinach, raw: 12% of the DV in 1 cup (25 g)
- Spinach, cooked: 39% of the DV in a 1/2 cup (90 g)
- Collard greens, cooked: 17% of the DV in 1/2 cup (65 g)
- Turnip greens, cooked: 21% of the DV in 1/2 cup (73 g)

3) Organ meats



Though not especially popular, organ meats — especially liver — are packed with B vitamins. For example, 3.5-oz (100-g) of beef liver contains:

- Riboflavin (B2): 263% of the DV
- Niacin (B3): 109% of the DV
- Pantothenic acid (B5): 139% of the DV
- Biotin (B7): 139% of the DV

4) Eggs



One large egg contains 35% of the DV for biotin distributed between the yolk and white. In fact, eggs are one of the top sources of biotin — only the liver contains more. Eggs also contain smaller amounts of other B vitamins. One large (50-g) cooked egg contains:

- Riboflavin (B2): 20% of the DV
- Biotin (B7): 35% of the DV
- Cobalamin (B12): 23% of the DV

5) Milk



Each cup, or 240 milliliters (ml), of milk, provides 26% of the DV for riboflavin, as well as smaller amounts of other B vitamins, including:

- Thiamine (B1): 9% of the DV
- Riboflavin (B2): 32% of the DV
- Pantothenic acid (B5): 18% of the DV
- Cobalamin (B12): 46% of the DV

DV = Daily Value
IU = International Unit

CALCIUM

Calcium is an essential mineral for maintaining good health. It plays a crucial role in building and maintaining strong bones and teeth, and also contributes to the normal functioning of many other systems in the body.

- Bone Health: Calcium is necessary for the formation and maintenance of bones. It helps to create a strong bone matrix, which gives bones their strength and structure. Calcium also helps to prevent bone loss and reduces the risk of osteoporosis, a condition that causes bones to become weak and brittle.
- Muscle Function: Calcium is involved in muscle contraction and relaxation. When calcium levels are low, the body may experience muscle cramps, weakness, and fatigue.
- Nerve Function: Calcium plays a role in nerve impulse transmission, which is essential for the communication between the brain and other parts of the body.
- Blood Clotting: Calcium is necessary for blood clotting. It helps to activate enzymes that are involved in the blood coagulation process, which is necessary to stop bleeding after an injury.
- Hormone Regulation: Calcium helps to regulate the activity of many hormones in the body, including those that control blood pressure, and the release of insulin.



The 5 best foods for calcium, that should be on your shopping list:

1) Seeds 	Seeds are tiny nutritional powerhouses, and many are high in calcium, including poppy, sesame, celery, and chia seeds. For instance, 1 tablespoon (9 grams) of poppy seeds packs 127 mg of calcium or 10% of the recommended Daily Value (DV). Sesame seeds contain 7% of the DV for calcium in 1 tablespoon (9 grams), plus other minerals, including copper, iron, and manganese.
2) Cheese 	Most cheeses are excellent sources of calcium. Parmesan cheese has the most, with 242 mg — or 19% of the DV — per ounce (28 grams). Softer cheeses tend to have less. For instance, 1 ounce (28 grams) of Brie only delivers 52 mg or 4% of the DV. As a bonus, your body absorbs the calcium in dairy products more easily than that from plant sources.
3) Sardines and salmon 	Sardines and canned salmon are loaded with calcium, thanks to their edible bones. A 3.75-ounce (92-gram) can of sardines packs 27% of the DV, and 3 ounces (85 grams) of canned salmon with bones has 19%.
4) Beans and lentils 	Beans and lentils are high in fiber, protein, and micronutrients. Some varieties also have decent amounts of calcium, including winged beans, which supply 244 mg, or 19% of the DV, in a single cooked cup (172 grams). White beans are also a good source, with 1 cup (179 grams) of cooked white beans providing 12% of the DV. Other varieties of beans and lentils have less, ranging from around 3-4% of the DV per cup (175 grams).
5) Yogurt 	Yogurt is an excellent source of calcium. Many types of yogurt are also rich in probiotics, a type of beneficial bacteria that can promote immune function, improve heart health, and enhance nutrient absorption. One cup (245 grams) of plain yogurt contains 23% of the DV for calcium, as well as a hearty dose of phosphorus, potassium, and vitamins B2 and B12.

DV = Daily Value
IU = International Unit

How much calcium do we need?

It's important to note that these are general guidelines, and individual needs may vary depending on factors such as age, gender, body size, and overall health status. Additionally, calcium absorption can be affected by other nutrients, such as vitamin D and magnesium, as well as certain medications and health conditions.

Consulting with a healthcare professional can help determine the appropriate amount of calcium needed for each individual.

Life stage	RDA
Birth to 6 months	200 mg
Infants 7–12 months	260 mg
Children 1–3 years	700 mg
Children 4–8 years	1000 mg
Children 9–13 years	1300 mg
Teens 14–18 years (boys)	1300 mg
Teens 14–18 years (girls)	1300 mg
Adults (men)	1000 mg
Adults (women)	1000 mg
Pregnant women	1000-1300 mg
Breastfeeding women	1000-1300 mg



IRON

Iron is an essential mineral for good health. It is an important component of hemoglobin, a protein found in red blood cells that carries oxygen throughout the body. Iron is also involved in many other physiological processes, including energy production, immune function, and cognitive development.

Some of the ways that iron supports good health are:

- Oxygen transport: Iron is a key component of hemoglobin, which is responsible for transporting oxygen from the lungs to the rest of the body. Iron deficiency can lead to anemia, a condition in which there are not enough red blood cells to carry adequate oxygen to the body's tissues.
- Energy production: Iron is also involved in the production of energy in cells. It helps to transport electrons within cells and is a key component of enzymes that are involved in energy production.
- Immune function: Iron plays a role in the immune system by helping to produce white blood cells, which are involved in fighting off infections and diseases.
- Cognitive development: Iron is important for cognitive development in children. Iron deficiency during critical periods of brain development can lead to long-lasting cognitive impairments.



The 5 best foods for iron, that should be on your shopping list:

1) Shellfish 	Shellfish is tasty and nutritious. All shellfish is high in iron, but clams, oysters, and mussels are particularly good sources. For instance, a 3.5-ounce (100-gram) serving of clams may contain up to 3 mg of iron, which is 17% of the DV. The iron in shellfish is heme iron, which your body absorbs more easily than the non-heme iron found in plants.
2) Spinach 	Spinach provides many health benefits but very few calories. About 3.5 ounces (100 grams) of raw spinach contain 2.7 mg of iron or 15% of the DV. Although this is non-heme iron, which isn't absorbed very well, spinach is also rich in vitamin C. This is important since vitamin C significantly boosts iron absorption.
3) Legumes 	Legumes are a great source of iron, especially for vegetarians. One cup (198 grams) of cooked lentils contains 6.6 mg, which is 37% of the DV. Beans like black beans, navy beans, and kidney beans can all help easily bump up your iron intake. In fact, a half-cup (86-gram) serving of cooked black beans provides around 1.8 milligrams of iron, or 10% of the DV.
4) Red meat 	A 3.5-ounce (100-gram) serving of ground beef contains 2.7 mg of iron, which is 15% of the DV. Meat is also rich in protein, zinc, selenium, and several B vitamins. Researchers have suggested that iron deficiency may be less likely in people who eat meat, poultry, and fish on a regular basis.
5) Pumpkin seeds 	A 1-ounce (28-gram) serving of pumpkin seeds contains 2.5 mg of iron, which is 14% of the DV. In addition, pumpkin seeds are a good source of vitamin K, zinc, and manganese. They're also among the best sources of magnesium, which many people are low in

DV = Daily Value
IU = International Unit

How much iron do we need?

Vegetarians and vegans may need to take extra care to ensure they get enough iron in their diet, as plant-based sources of iron are less easily absorbed by the body. Iron supplements are also available, but it's important to talk to a healthcare professional before taking them, as excessive intake of iron can lead to health problems.

Life stage	RDA
Birth to 6 months	0.27 mg
Infants 7–12 months	11 mg
Children 1–3 years	7 mg
Children 4–8 years	19 mg
Children 9–13 years	8 mg
Teens 14–18 years (boys)	11 mg
Teens 14–18 years (girls)	15 mg
Adults (men)	8 mg
Adults (women)	18 mg
Pregnant women	27 mg
Breastfeeding women	9 mg



MAGNESIUM

Magnesium is an essential mineral that is involved in a wide range of bodily functions. It is required for the proper functioning of the nervous, muscular, and cardiovascular systems, as well as for bone health and the synthesis of DNA and proteins. Magnesium is also involved in the regulation of blood sugar levels, blood pressure, and the production of energy.

Magnesium can be found in a variety of foods, including green leafy vegetables, nuts, seeds, whole grains, and legumes. However, the amount of magnesium that is absorbed by the body can vary depending on factors such as the presence of other nutrients and the form of the magnesium. For example, magnesium is more easily absorbed from sources such as magnesium citrate or magnesium glycinate, rather than from sources such as magnesium oxide.

A deficiency of magnesium can lead to a variety of health problems, including muscle cramps, fatigue, and a weakened immune system. Severe magnesium deficiency can also cause heart arrhythmias and seizures. However, excessive intake of magnesium can also be harmful, leading to diarrhea and nausea.



The 5 best foods for magnesium, that should be on your shopping list:

1) Dark chocolate 	Dark chocolate is as nutritious as it is flavorful. It's very rich in magnesium, with 65 mg in a 1-ounce (oz.), or 28-gram (g), serving, which is about 15% of the DV. Dark chocolate is also high in iron, copper, and manganese and contains prebiotic fiber that can help feed the beneficial bacteria in the gut.
2) Avocado 	Avocado is an incredibly nutritious fruit and a tasty source of magnesium. One medium avocado provides 58 mg of magnesium, 14% of the DV. Avocados are also high in potassium, B vitamins, and vitamin K. Plus, unlike most fruits, they're high in fat — especially heart-healthy monounsaturated fat.
3) Legumes 	They're very rich in many different nutrients, including magnesium. For instance, a 1-cup (172-g) serving of cooked black beans contains an impressive 120 mg of magnesium, which is 29% of the DV. Legumes are also high in potassium and iron and a major source of protein for vegetarians.
4) Tofu 	Tofu is a staple food in vegetarian diets due to its high protein content. Made by pressing soybean milk into soft white curds, it's also known as bean curd. A 3.5-oz. (100-g) serving contains 35 mg of magnesium, which is 8% of the DV.
5) Seeds 	Many varieties — including flax, pumpkin, and chia seeds — also contain high amounts of magnesium. Pumpkin seeds are a particularly good source, with 168 mg in a 1-oz. (28-g) serving. This amounts to a whopping 40% of the DV for magnesium. In addition, seeds are rich in iron, monounsaturated fat, and omega-3 fatty acids

DV = Daily Value
IU = International Unit

How much magnesium do we need?

Even though supplementation can correct a low magnesium blood level, you can also increase your level naturally through diet.
The recommended daily amount of magnesium for adult females is 320 mg to 360 mg, and 400 mg to 420 mg for adult males, according to the NIH.

It's important to note that these are general recommendations, and individual needs may vary based on factors such as overall health and diet.

Tap water, mineral water, and bottled water are also sources of magnesium, although magnesium levels can vary depending on the water source.

Life stage	RDA
Birth to 6 months	30 mg
Infants 7–12 months	75 mg
Children 1–3 years	80 mg
Children 4–8 years	1300 mg
Children 9–13 years	240 mg
Teens 14–18 years (boys)	410 mg
Teens 14–18 years (girls)	360 mg
Adults (men)	400-420 mg
Adults (women)	320-360 mg
Pregnant women	350-360 mg
Breastfeeding women	310-360 mg



ZINC

Zinc is an essential mineral that is important for a variety of bodily functions. It is required for the proper growth and development of the body, as well as for the immune system to function effectively. Zinc is involved in the metabolism of carbohydrates, protein, and fat, and it also plays a role in wound healing, DNA synthesis, and cell division.

Zinc can be found in a variety of foods, including meat, poultry, seafood, whole grains, beans, nuts, and dairy products. However, the amount of zinc that is absorbed by the body can vary depending on factors such as the form of the zinc and the presence of other nutrients. For example, the zinc in animal products is generally more easily absorbed than the zinc in plant-based foods.

A deficiency of zinc can lead to a variety of health problems, including delayed wound healing, decreased sense of taste and smell, and a weakened immune system. Severe zinc deficiency can also cause growth retardation, hair loss, and skin lesions. However, excessive intake of zinc can also be harmful, leading to nausea, vomiting, and diarrhea.



The 5 best foods for Zinc, that should be on your shopping list:

1) Oysters



Oysters are a particularly rich source of zinc, with about 74 mg of zinc per 100 grams. This is more than seven times the recommended daily intake of zinc for adults. Oysters also contain other important nutrients, such as vitamin B12, copper, and iron.

2) Beef



Beef is a good source of zinc, with about 7 mg per 100 grams. This is about 60% of the recommended daily intake for adults. Beef is also a good source of protein, which is essential for building and repairing tissues in the body.

3) Pumpkin seeds



Pumpkin seeds are a great plant-based source of zinc, with about 7 mg per 100 grams. This is about 60% of the recommended daily intake for adults. In addition to zinc, pumpkin seeds are also high in other important nutrients, including magnesium (which is important for bone health and nerve function), phosphorus, and antioxidants.

4) Chickpeas



Chickpeas are versatile legumes that are high in zinc, with about 1 mg per 100 grams. This is about 10% of the recommended daily intake for adults. Chickpeas are also a good source of protein, and fiber (which helps to promote feelings of fullness and can aid in digestion).

5) Spinach



Spinach is a leafy green vegetable that is a good source of zinc, with about 0.5 mg per 100 grams. It is also high in other important nutrients, including iron, vitamin C, and vitamin K.

DV = Daily Value
IU = International Unit

How much zinc do we need?

The body's ability to absorb zinc changes as we age, so the amount of zinc required each day depends on your age. The range on the right side, developed by the Food and Nutrition Board (FNB) at the Institute of Medicine of the National Academies, can help you find your recommended dietary allowance (RDA).

Most people who consume a varied diet rich in whole foods get adequate zinc intake in their diet. However, zinc is available in most multivitamins and as a stand-alone supplement.

Excess zinc intake and toxicity may cause reduced immunity and other severe complications. Consult your doctor or dietitian to determine deficiency and whether a supplement is needed.

**Vegetarians and vegans may require 50% more of the recommended daily allowance (RDA) for zinc than meat-eaters.*

Life stage	RDA
Birth to 6 months	2 mg
Infants 7–12 months	3 mg
Children 1–3 years	3 mg
Children 4–8 years	5 mg
Children 9–13 years	8 mg
Teens 14–18 years (boys)	11 mg
Teens 14–18 years (girls)	9 mg
Adults (men)	11 mg
Adults (women)	8 mg
Pregnant women	11 mg
Breastfeeding women	12 mg



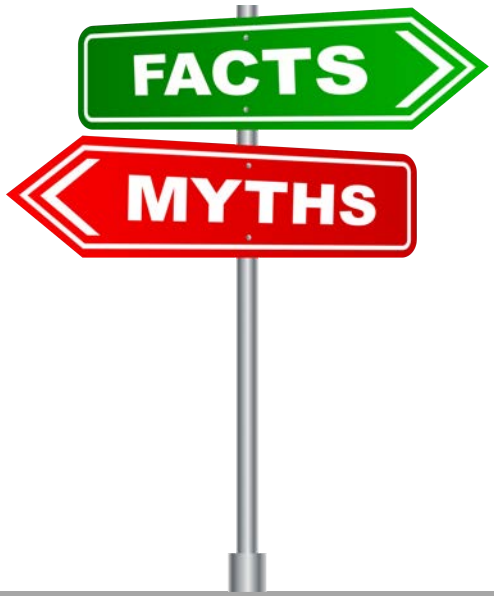
COMMON MYTHS AND MISCONCEPTIONS ABOUT VITAMINS AND MINERALS

More is always better

When it comes to vitamins, more is not always better. In fact, more can sometimes be dangerous. As vitamin and mineral supplements are available without a prescription, people can be forgiven for assuming that they are safe at any dosage.

Large dosages of some vitamins can hamper the body's finely tuned systems, though. For instance, according to the American Cancer Society: "Too much vitamin C can interfere with the body's ability to absorb copper, a metal that's needed by the body. Too much phosphorous can inhibit the body's absorption of calcium. The body cannot get rid of large doses of vitamins A, D, and K, and these can reach toxic levels when too much is taken."

Additionally, too much vitamin C or calcium might cause diarrhea and stomach pain. Taking too much vitamin D over long periods can cause calcium to build up in the body, which is called hypercalcemia. Hypercalcemia can weaken bones and damage the heart and kidneys.



Vitamin and mineral supplements protect heart health

The idea that taking vitamin and mineral supplements will protect our hearts is reassuring. However, a large review and meta-analysis published in 2018 could find no significant benefit. Overall, the authors conclude:

"In general, the data on the popular supplements (multivitamins, vitamin D, calcium, and vitamin C) show no consistent benefit for the prevention of [cardiovascular disease, myocardial infarction, or stroke], nor was there a benefit for all-cause mortality to support their continued use."

Although they did find that "Folic acid alone and B vitamins with folic acid, B6, and B12 reduced stroke," overall, these effects were small.

With that said, a 2019 meta-analysis did identify links between folic acid supplementation and reduced stroke risk in people with cardiovascular risk. And the results of another study "indicated a 10% lower risk of stroke and a 4% lower risk of overall [cardiovascular disease] with folic acid supplementation."

Vitamin C prevents a cold

Although there is a grain of truth to this slice of "common knowledge," the evidence that vitamin C can prevent a cold is weak.

For instance, a 2013 Cochrane review dove into existing evidence. The authors' objective was to "find out whether vitamin C reduces the incidence, the duration, or severity of the common cold when used either as a continuous regular supplementation every day or as a therapy at the onset of cold symptoms."

The scientists found that vitamin C supplementation did not prevent the common cold in the general population. However, they did conclude that it reduced the severity of symptoms and the duration of the cold.

They also concluded that it "may be useful for people exposed to brief periods of severe physical exercise," such as marathon runners.

Sources & Further Reading

Vitamin D deficiency: A single centre analysis of patients from 136 countries.
https://pubmed.ncbi.nlm.nih.gov/26877203/

Comparative analysis of nutritional guidelines for vitamin D.
https://pubmed.ncbi.nlm.nih.gov/28387318/

William B. Grant, PhD, and Michael F. Holick, PhD, MD. Benefits and Requirements of Vitamin D for Optimal Health: A Review
https://pubmed.ncbi.nlm.nih.gov/15989379/
https://altmedrev.com/wp-content/uploads/2019/02/v10-2-94.pdf

Lips P, Hosking D, Lippuner K, Norquist JM, Wehren L, Maalouf G, et al. The prevalence of vitamin D inadequacy amongst women with osteoporosis: An international epidemiological investigation.
https://onlinelibrary.wiley.com/doi/full/10.1111/j.1365-2796.2006.01685.x

H F DeLuca. The control of calcium and phosphorus metabolism by the vitamin D endocrine system.
https://pubmed.ncbi.nlm.nih.gov/7015957/

H F DeLuca, H K Schnoes. Vitamin D: recent advances.
https://pubmed.ncbi.nlm.nih.gov/6311080/

Joan M. Lappe, PhD, RN, FAAN. The Role of Vitamin D in Human Health: A Paradigm Shift.
https://journals.sagepub.com/doi/pdf/10.1177/1533210110392952

Li X, Liu Y, Zheng Y, Wang P, Zhang Y. The effect of vitamin D supplementation on glycemic control in type 2 diabetes patients: A systematic review and meta-analysis.
https://pubmed.ncbi.nlm.nih.gov/29562681/

Human Nutrition and Health (IB, AF, FFR), and Nutrition Science & Advocacy (PW, ME), DSM Nutritional Products Ltd., Basel, Switzerland. Vitamin D: a critical and essential micronutrient for human health.
https://www.frontiersin.org/articles/10.3389/fphys.2014.00248/full

Mousa A, Naderpoor N, Teede H, Scragg R, de Courten, B. Vitamin D supplementation for improvement of chronic low-grade inflammation in patients with type 2 diabetes: A systematic review and meta-analysis of randomized controlled trials.
https://pubmed.ncbi.nlm.nih.gov/29490085/

A Review of Mushrooms as a Potential Source of Dietary Vitamin D.
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6213178/

United States Department of Agriculture (USDA) Food Composition Database.
https://fdc.nal.usda.gov/index.html

10 Magnesium-Rich Foods That Are Super Healthy
https://www.healthline.com/nutrition/10-foods-high-in-magnesium

The 10 Best Foods That Are High in Zinc
https://www.healthline.com/nutrition/best-foods-high-in-zinc#The-10-Best-Foods-That-Are-High-in-Zinc

Dietary Vitamin C in Human Health
https://pubmed.ncbi.nlm.nih.gov/29477224/

https://fdc.nal.usda.gov/

Daily Value on the New Nutrition and Supplement Facts Labels
https://www.fda.gov/food/new-nutrition-facts-label/daily-value-new-nutrition-and-supplement-facts-labels

Cocoa Polyphenols and Gut Microbiota Interplay: Bioavailability, Prebiotic Effect, and Impact on Human Health
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7400387/

The Odyssey of Bioactive Compounds in Avocado (Persea americana) and Their Health Benefits
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6826385/

Dietary Protein and Amino Acids in Vegetarian Diets—A Review
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6893534/

Iron Rich Foods
https://www.webmd.com/diet/iron-rich-foods

Biotin
https://ods.od.nih.gov/factsheets/Biotin-HealthProfessional/

Dietary Intake and Food Sources of Niacin, Riboflavin, Thiamin and Vitamin B_6 in a Representative Sample of the Spanish Population. The Anthropometry, Intake, and Energy Balance in Spain (ANIBES) Study
https://pubmed.ncbi.nlm.nih.gov/29966236/

Vitamin K: the effect on health beyond coagulation - an overview
https://pubmed.ncbi.nlm.nih.gov/22489224/

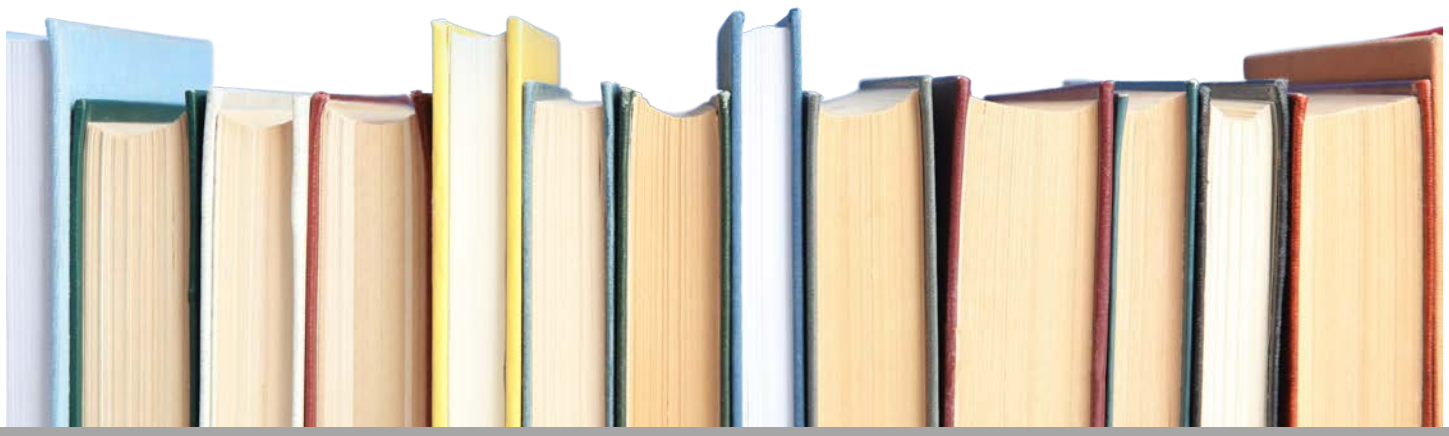
Vitamin K Intake and bone mineral density in women and men
https://pubmed.ncbi.nlm.nih.gov/12540415/

Dietary Supplements
https://www.cancer.org/treatment/treatments-and-side-effects/treatment-types/complementary-and-integrative-medicine/dietary-supplements/intro.html

The effect of folic acid in patients with cardiovascular disease: A systematic review and meta-analysis
https://pubmed.ncbi.nlm.nih.gov/31517834/

Folic Acid Supplementation and the Risk of Cardiovascular Diseases: A Meta-Analysis of Randomized Controlled Trials
https://pubmed.ncbi.nlm.nih.gov/27528407/

Vitamin C for preventing and treating the common cold
https://www.cochranelibrary.com/cdsr/doi/10.1002/14651858.CD000980.pub4/full





ENCHANTED
MEDICAL AESTHETICS

276 N. Nova Rd
Ormond Beach, Fl 32174
(386) 317-1449

Dr. Bridget Martin
DNP, APRN, NP-C

LET'S CONNECT

enchanted_aesthetics

Enchanted Medical Aesthetics

ENCHANTED MEDICAL AESTHETICS

EnchantedMedicalAesthetics.com